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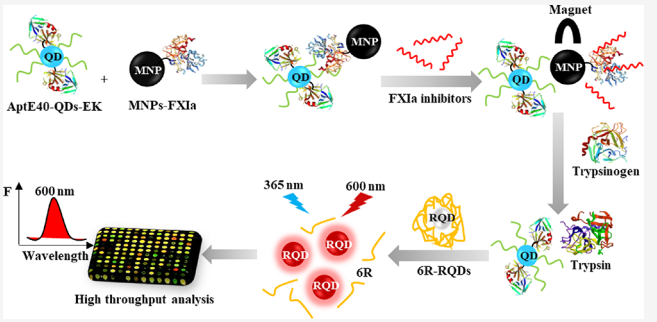
Development of an Ultrasensitive Competitive Double-Enzyme Cascade Fluorescence Signal Amplification Method for Factor XIa Inhibitor Screening

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ABSTRACT: Coagulation factor XIa (FXIa) is associated with a low risk of bleeding and has been identified as an effective and safe target for the development of novel anticoagulant drugs. In this study, we established an ultrasensitive competitive dual-enzyme cascade signal amplification method for the quantitative analysis and screening of FXIa inhibitors. Due to the specific recognition of FXIa's active site by the aptamer AptE40, the AptE40-QDs-EK recognition probe modified with enterokinase (EK) and the aptamer AptE40, was attached to the MNPs-FXIa capture probe. When FXIa inhibitor was present, it competed with AptE40 for binding to FXIa, resulting in the detachment of AptE40-QDs-EK from MNPs-FXIa. After magnetic separation, the enterokinase of AptE40-QDs-EK in the supernatant hydrolyzed N-terminal hexapeptide of trypsinogen, leading to the production of a large amount of trypsin as part of the first-stage signal cascade amplification. Next, trypsin could hydrolyze the hexameric arginine peptide (RRRRRR, R6), leading to the dissociation of RQDs from the R6-RQDs signal probe; this resulted in a dramatic increase in the fluorescence intensity of the supernatant as the second-stage signal cascade was amplified. The feasibility of the method was investigated using the FXIa inhibitor aptamer FELIAP as a positive model drug. Furthermore, the method was applied to screen the FXIa inhibitors in *Eupolyphaga sinensis* Walker. Two fractions with more active anticoagulated ingredients were successfully identified and validated via the conventional method, and the results were consistent. The established method provides a key technique for the sensitive detection, high-throughput analysis, and screening of the FXIa inhibitors.



INTRODUCTION

According to the World Health Organization (WHO), cardiovascular diseases account for more than 43% of annual deaths worldwide, reaching a 17.3 million cases, and this number is projected to surpass 23.6 million by 2030.[1−3] Currently, commonly used anticoagulation drugs (such as heparin and warfarin) exhibit favorable clinical efficacy. However, they pose potential bleeding risks because of their interference with the intricate balance between procoagulation and anticoagulation in the body.[4−6] Clinical evidence indicates that FXI/FXIa play a pivotal role in the intrinsic coagulation pathway by activating FIX to FIXa and lowering the risk of bleeding through the common coagulation pathway.[7−9] Consequently, many studies have focused on developing anticoagulants that target FXI/FXIa as potential novel drugs.[10,11] Currently, several FXIa inhibitors are at various stages of research and are derived mainly from natural products and their derivatives, such as chemically synthesized small molecule compounds, peptides, and antisense oligonucleotides.[12] Clavatadine A was able to bind irreversibly to the active site of FXIa and exert an inhibitory effect (IC_{50} = 1.3 μmol/L). Al-Horani et al.[13] discovered that sulfated pentaglucopyide (SPGG) exhibited potent allosteric inhibition of FXIa by selectively targeting multiple anionic allosteric sites on the enzyme.[14] Argade et al. screened a library of molecules from natural product fragments with diverse backbone structures that exhibited inhibitory effects on FXIa.[15] In the above research process, the components were very complex and had large interferences. Moreover, coagulation is a complex disease network system involving abnormal activation of coagulation factors, platelet aggregation, vascular endothelial injury, increased blood viscosity. Therefore, in the early stage of screening, rapid screening and activity evaluation of the trace anticoagulant components in complex matrices are difficult. A

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sensitive, rapid, specific method to screen and evaluate the activity of novel FXIa inhibitors needs to be established.

Lee et al. introduced a novel technique known as cascade enzyme-linked immunosorbent assay (Cascade ELISA or CELISA), and this assay successfully enabled ultrasensitive detection of alpha-fetoprotein (AFP) in biological samples.[16] Compared with the traditional ELISA method, this method has extremely high signal amplification efficiency and sensitivity. At present, CELISA technology is expected to become an alternative to ELISA.[17] On the one hand, the dual-enzyme signal amplification system can greatly enhance the detection signal. On the other hand, the use of nanoparticles instead of microplates as solid-phase carriers can result in a larger effective area and achieve higher capture efficiency. Previous studies have applied nanoparticles, such as quantum dots, gold nanoparticles, silica nanoparticles and other materials, to amplify detection signals. Owing to their controllable size, convenient synthesis, large specific surface area and enhanced signal, nanoparticles provide a good microscopic platform for signal amplification technology. With the development of biocompatible modified materials, the functionalized modification technology for nanoparticles has become increasingly mature. The functionalized nanoparticles are formed by electrostatic or covalent binding of biological macromolecules, small molecular compounds or drug molecules, and these nanoparticles have achieved a wide range of applications in biomedicine.[18−20]

In this work, we established an ultrasensitive fluorescence screening and quantitative method for detecting FXIa inhibitors through a competitive double-enzyme cascade signal amplification method (Figure 1). The MNPs-FXIa capture probe and AptE40-QDs-EK recognition probe were designed to screen the FXIa inhibitor in complexes. The enterokinase and trypsinogen formed the double-enzyme cascade to amplify the fluorescence signal of RQDs. Finally, the degree of fluorescence signal recovery correlates with the FXIa inhibitory content. The feasibility of this method was investigated using the FXIa inhibitor FELIAP as a positive model drug, which has been previously reported. Furthermore, we successfully applied this method to screen FXIa inhibitors in crude protein extracts from *Eupolyphaga sinensis* Walker, resulting in the validation of two highly active fractions confirmed by conventional anticoagulant analysis. This work has significant application potential in novel anticoagulant pharmaceutical research and for screening the FXIa inhibitor in the complexes.

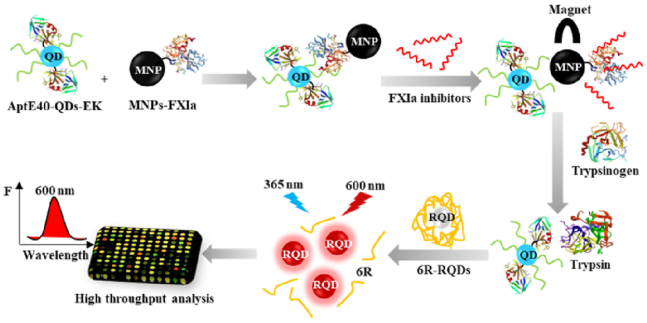


Figure 1. Schematic diagram of the ultrasensitive competitive double-enzyme cascade fluorescence signal amplification method for screening FXIa inhibitors in complexes.

EXPERIMENTAL SECTION

AptE40-QDs-EK Recognition Probe. In a N_2-sealed three-necked flask, sodium borohydride ($NaBH_4$, 0.4 mmol), selenium powder (Se, 0.1 mmol) and redistilled water (500 μL) were magnetically stirred for approximately 40 min. After the complete reaction of the black selenium powder, an ice bath was applied for 10 min. L-glutathione (GSH, 0.4 mmol) was added to the Zn $(Ac)_2{\cdot}2H_2O$ aqueous solution (4 mM, pH 10.5) adjusted with a NaOH solution (1 M). The mixture was magnetically stirred under a N_2 atmosphere for 30 min. The prepared NaHSe solution (1 mM) was added and stirred for another 30 min. The mixture was refluxed at 100 °C for 90 min to obtain a transparent and colorless QDs solution (1 mM). The QDs solution was subsequently concentrated to approximately one-third of its original volume and further stirred with anhydrous ethanol (1.5 times the volume). After centrifugation at 16500 × *g* for 30 min, the white precipitate was vacuum-dried at 60 °C. The QDs (10 mg/mL, 100 μL), AptE40 (2 nmol), recombinant human enterokinase (EK, 5 μg), and EDC (10 mg) were incubated in HEPES buffer (pH 7.5, 25 mM) at room temperature for 2 h, and the resulting precipitate was then collected by centrifugation at 16500 × *g* for 5 min. Subsequently, the QDs capture probe (AptE40-QDs-EK) cofunctionalized by enterokinase and AptE40 was washed three times with Tris-HCl buffer (pH 8.0, 25 mM), redispersed in 200 μL HEPES buffer (25 mM), and stored at 4 °C.

MNPs-FXIa Capture Probe. First, the MNPs were washed three times with the coupling buffer (25 mM HEPES, pH 7.5, containing 0.1% Tween-20). The MNPs (1 mL, 5 mg/mL), FXIa (0.1 mL, 5 mg/mL), and EDC (50 mg) were subsequently allowed to react at room temperature for 4 h. After magnetic separation, the MNPs-FXIa capture probes were washed three times with coupling buffer, redispersed in Tris-HCl buffer (pH 8.0, 25 mM, containing 0.1% Tween-20) and stored at 4 °C (5 mg/mL). The protein loading rate on the MNPs was calculated by using the BCA protein content assay kit.

R6-RQD Signal Probe. $CdCl_2{\bullet}2.5H_2O$ (0.8 mmol) and reduced glutathione (GSH, 1.6 mmol) were dissolved in ultrapure water (100 mL) and adjusted to pH 10.0 using NaOH solution (1 M) for the preparation of the cadmium precursor. The zinc precursor solution was obtained by adding $ZnSO_4{\bullet}7H_2O$ (1.26 mmol) and GSH (1.26 mmol) to ultrapure water (100 mL, pH 10.0). The powder (0.3 mmol) and $NaBH_4$ (1.8 mmol) were added to ultrapure water (3 mL) and stirred at room temperature for 65 min. Then, the freshly prepared NaHTe solution was rapidly injected into the previously prepared cadmium precursor solution under vigorous stirring and refluxed for 25 min. Subsequently, the zinc precursor solution was added and refluxed for an additional 15 min. Rotary evaporation was used to reduce its volume to one-third of the original amount, and then three times the volume of isopropanol was added. The brick-red solid RQDs were obtained by centrifugation at 2500 × *g* for 10 min and vacuum drying at 40 °C. To prepare the R6-RQDs signal probe, RQDs powder was dissolved in Tris-HCl buffer (pH 8.0, 25 mM, 150 μg/mL) and vortexed with an equal volume of hexameric arginine peptide solutions (R6, 50 μg/mL). The resulting precipitate was obtained by centrifugation at 16500 × *g* for 5 min, washed three times with Tris-HCl buffer (pH 8.0, 25 mM) and stored at 4 °C.

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Enterokinase Mediated Activation of Trypsinogen. The activation of trypsinogen by enterokinase was confirmed using sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS−PAGE). Enterokinase (50 μg/mL), trypsinogen (200 μg/mL), and trypsin (100 mg/mL) were prepared in Tris-HCl buffer (pH 8.0, 25 mM). Enterokinase (10 nM) and trypsinogen (400 μg/mL) were incubated for 30 min at room temperature. The sample (50 μL) was subsequently combined with 6 × protein (10 μL) loading buffer, boiled in a water bath for 5 min, incubated in an ice bath for another 5 min, and subsequently centrifuged at 9000 × *g* for 1 min. Electrophoresis was performed under constant current conditions. Initially, the constant current was set at 30 mA until the bromophenol blue entered the separation gel, and the current was then adjusted to 45 mA until reaching the lower edge of the gel. The gel was immersed in 0.25% Kaomas Brilliant Blue R250 at 25 °C for 2 h and then decolorized with methanol: water: glacial acetic acid (30:60:10, v/v/v) at 25 °C for 1 h. The resulting photographs were analyzed using ImageJ 2.0 software.

Trypsin-Mediated Degradation of Hexa-Arginine Peptide. To study the degradation of hexameric arginine (R6) by trypsin, we designed fluorescently labeled R6 peptide-modified SiO_2 (SiO_2-R6-DNS) for verification. First, aminated silica nanoparticles were synthesized using the reversed-phase microemulsion method. Cyclohexane (75 mL), triton X-100 (17.7 mL, 30.83 mM), *n*-hexanol (18 mL), ultrapure water (0.48 mL), and ammonia solution (2.4 mL, 25−28 wt %) were vigorously stirred for 30 min. Then, TEOS (4.51 mM) and APTES (4.27 mM) were added, and the mixture was further stirred for 72 h in the dark. The resulting precipitate was collected by centrifugation at 2500 × *g* for 5 min, washed three times with anhydrous ethanol, and dried to obtain the aminated silica nanoparticles (SiO_2). SiO_2 (20 mg), N-terminal dansyl chloride-labeled hexa-arginine peptide (DNS-R6, 20 mg), EDC (200 mg), and HEPES buffer (pH 7.5, 25 mM) were stirred at room temperature in the dark for 1 h. The precipitate was subsequently collected by centrifugation at 16,500 × *g* for 5 min, washed three times with HEPES buffer, redispersed in Tris-HCl buffer (pH 8.0, 25 mM), and stored at 4 °C. For the validation assay, SiO_2-R6-DNS (2 mg/mL) and trypsin (1 mg/mL) were incubated in the dark at 37 °C for various durations (0, 15, 30, 60, 90, 120, and 150 min) and then centrifuged at 16500 × *g* for 5 min. The fluorescence intensity at 535 nm was measured with an excitation wavelength of 310 nm.

Trypsin Activity Inhibition Assay. We inhibited trypsin activity to further validate the degradation of R6 by trypsin. Compared with that of untreated trypsin, the recovery of R6-RQDs fluorescence was attributed to the enzymatic activity of trypsin. In general, trypsin activity is inhibited through heating or the use of inhibitors. In this study, a trypsin solution (20 μg/mL) was boiled for 15 min in a water bath and then cooled to room temperature. A serine protease inhibitor, phenylmethylsulfonyl fluoride (PMSF, 0, 0.4, 0.8, 1.6, 2.0 mM), was used to inhibit trypsin activity. The inactivated or activated trypsin and R6-RQDs (75 μg/mL) were incubated at 25 °C for 10 min. Then, the fluorescence intensity at 600 nm was measured using excitation wavelength of 365 nm.

Feasibility Analysis. The quenching phenomenon of the R6 peptide on the RQDs was investigated by thoroughly vortexing R6 with different concentrations (0, 10, 20, 30, 40, 50, 60, and 70 μg/mL) of RQDs (150 μg/mL). The fluorescence intensity was then obtained at a wavelength of 600 nm with an excitation wavelength of 365 nm. To examine the interaction between R6-RQDs and trypsin, various concentrations (0, 5, 10, 15, and 20 μg/mL) of trypsin were incubated with R6-RQDs (75 μg/mL) for 10 min at 25 °C. The recovery of the fluorescence intensity of the R6-RQDs induced by trypsin was evaluated by measuring the fluorescence intensity at 600 nm. Enterokinase at different concentrations (0, 50, 100, 150, 200, 300, 400, and 500 μg/mL) and trypsinogen (0.1 mg/mL) were incubated at 25 °C for 2 h. The R6-RQDs (75 μg/mL) were subsequently added to the above mixture and further incubated at 25 °C for 5 min. The fluorescence recovery of R6-RQDs induced by enterokinase-activated trypsin was evaluated at 600 nm with an excitation wavelength of 365 nm.

FXIa Inhibitor Detection. MNPs-FXIa (5 mg/mL) and AptE40-QDs-EK (5 mg/mL) were incubated in Tris-HCl buffer (containing 1 M NaCl, 0.1 M, pH 7.4) at room temperature for 30 min. After magnetic separation, the precipitate was washed four times with Tris-HCl buffer (pH 8.0, 25 mM). Finally, the precipitate was redispersed in Tris-HCl buffer to prepare the MNPs-FXIa-AptE40-QDs-EK complex and stored at 4 °C for future use. For the detection of FELIAP, MNPs-FXIa-AptE40-QDs-EK and various concentrations of FELIAP solution (1, 10, 100, 200, 300, 400 nM) were incubated at room temperature for 20 min. After magnetic separation, trypsinogen (0.1 mg/mL) along with freshly prepared R6-RQDs (75 μg/mL) were added. The mixture was incubated at room temperature for 10 min before the fluorescence intensity was measured at 600 nm using an excitation wavelength of 365 nm.

FXIa Inhibitor Screening. The *Eupolyphaga sinensis Walker* (2 g) was pulverized and hydrolyzed with pepsin (2%) and trypsin (4%) successively for 3 h at 37 °C. The resulting crude protein extract was filtered using Sephadex TM G-75 chromatography (2 × 50 cm), and the fractions obtained were lyophilized and collected for screening. FELIAP (1 μM) served as a positive control group, while ultrapure water was used as a negative control group. The MNPs-FXIa-AptE40-QDs-EK (10 μL) and the extract (1 μg/mL) or control samples were incubated at room temperature for 20 min. The supernatant was collected, and trypsinogen (0.1 mg/mL) with R6-RQDs (75 μg/mL) was added. After incubation for 10 min at room temperature, the fluorescence intensity at 600 nm was measured using an excitation wavelength of 365 nm. ΔF/Fn was employed as an indicator to assess the inhibitory efficacy against FXIa, where ΔF represents the difference between the fluorescence intensities at 600 nm observed in the experimental group and the corresponding calibration group respectively, and Fn is the corrected fluorescence intensity measured in the negative control group.

Anticoagulant Activity Validation. Conventional APTT and PT kits were employed for the validation of anticoagulant activity screening. Male Sprague−Dawley rats (180−200 g) were housed under conditions of constant temperature (24 ± 1 °C), relative humidity (50 ± 10%), and a 12 h light−dark cycle. All animal procedures adhered to the guidelines outlined in the National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publication No. 8023, revised in 1978) and received approval from the Animal Ethics Committee of China Pharmaceutical University (no. 2023−05−012). Blood was collected from the rats following anesthesia, and 3.8% sodium citrate was used as an

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anticoagulant. The collected blood was then centrifuged at 900 × *g* for 10 min to obtain platelet-poor plasma (PPP). For the APTT assay, a sample solution (10 mg/mL) was mixed with PPP and APTT reagents. The mixture was thoroughly mixed and incubated at 37 °C for 3 min, followed by the addition of $CaCl_2$. For the PT assay, a sample (10 mg/mL) was mixed with PPP and incubated at 37 °C for 3 min. The APTT and PT assays were determined using an SC40 semiautomatic coagulation analyzer.

RESULTS AND DISCUSSION

Characterization. Transmission electron microscopy (TEM) images of the QDs revealed a spherical morphology (Figure S1A) with an average particle size of 3.0 ± 0.5 nm (Figure S1B). The diffraction ring diameters were 0.255, 0.412, and 0.479 nm from the inner to outer rings, corresponding to radii squared of 0.0163, 0.0431, and 0.0583 nm^2, respectively (at a ratio of approximately 1:2.6:3.6). These results indicated the presence of typical face-centered cubic crystal structures with crystallographic indices of (111), (220), and (311) from the innermost to outermost rings. Furthermore, the results showed that smaller nanoparticle sizes provided larger specific surface areas for improved functionalized modifications, reducing spatial site resistance and enhancing binding efficiency. The UV absorption spectra of the QDs with AptE40-QDs-EK are presented in Figure S2. Both the QDs and AptE40-QDs-EK exhibited a distinct exciton absorption peak at 360 nm (Figure S2a,b). Furthermore, the presence of a new absorption peak near 260 nm for AptE40-QDs-EK (Figure S2c) indicated successful modification of the AptE40 aptamer on the surface of the QDs without affecting their optical properties. Notably, due to its proximity to the UV absorption wavelength of enterokinase at 280 nm, the specific absorbance peak related to enterokinase was not prominently visible in the spectra of AptE40-QDs-EK. The enterokinase modification of the QDs was verified by assessing the enterokinase activity of the AptE40-QDs-EK recognition probe. As shown in Figure 2A, upon coincubation of AptE40-QDs-EK with trypsinogen and R6-RQDs, AptE40-QDs-EK rapidly activated trypsinogen, resulting in the restoration of the RQDs fluorescence, which was similar to that of enterokinase (1 nM) activation. Conversely, the RQDs fluorescence did not recover in the negative control group. These findings clearly demonstrated the successful surface modification of the QDs with active enterokinase.

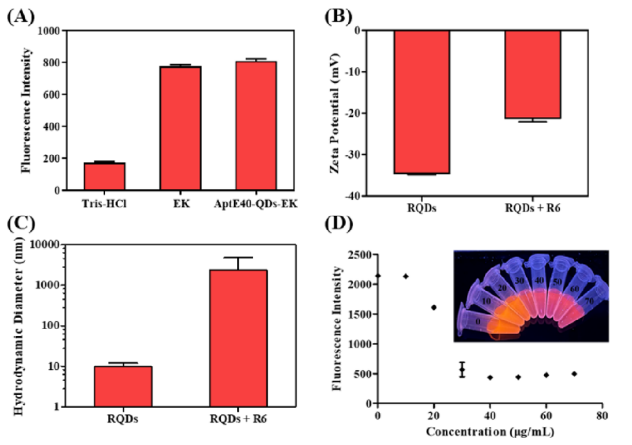


Figure 2. (A) Validation of the enterokinase activity of AptE40-QDs-EK. Changes in the zeta potential (B) and particle size (C) of the RQDs (1 mg/mL) induced by R6 peptide (0.3 mg/mL). (D) Fluorescence intensity changes of RQDs (150 μg/mL) with different concentrations (0, 10, 20, 30, 40, 50, 60, and 70 μg/mL) of the R6 peptide at 600 nm with an excitation wavelength of 365 nm (all insets are photographs at 365 nm under a UV lamp).

R6-RQDs Signal Probe. As shown in Figure S3A,B, the TEM images of the RQDs revealed a spherical morphology with an average particle size of 3.5 ± 0.5 nm. The diffraction ring diameters were 0.255, 0.412, and 0.479 nm from the inner ring to the outer ring, corresponding to radii squared values of 0.0163, 0.0431, and 0.0583 nm^2, respectively (1:2.6:3.6). These observations indicated the presence of typical face-centered cubic crystal structures with crystallographic indices of (111), (220), and (311) for the order sequence of the respective rings. The surface of the RQDs was stabilized by glutathione (pKa1 = 2.21) and 3-mercaptopropionic acid (pKa1 = 4.34). These carboxyl groups are typically ionized under neutral or weakly alkaline conditions, resulting in an abundance of negative charges on the RQD surface. This electrostatic repulsion between the QDs prevented aggregation and played a crucial role in maintaining their stability in solution. Arginine (Arg, R) is a prototypical basic amino acid, and its side chain guanidinium group becomes positively charged through protonation under weakly acidic, neutral, and even weakly basic conditions. Consequently, the R6 peptide is a linear oligopeptide with a substantial number of positive charges. In the presence of RQDs in solution, electrostatic forces facilitated the binding of the R6 peptide to the RQDs, resulting in a significant reduction in their negative charge. As shown in Figure 2B, the incorporation of the R6 peptide (0.3 mg/mL) and RQDs (1 mg/mL) resulted in a significant reduction in the zeta potential of the RQDs from −34.6 ± 0.3 to −21.3 ± 0.8 mV, thereby compromising their stability in aqueous solution and promoting aggregation due to weakened electrostatic repulsion. As shown in Figure 2C, the hydrated diameter of the QDs significantly increased from 9.97 ± 2.01 nm to 2.88 ± 1.67 μm; these results indicated their aggregation from a monodisperse state into larger particles. More importantly, this aggregation induced an evident change in fluorescence intensity. When RQDs were mixed with the R6 peptide at various concentrations (0, 10, 20, 30, 40, 50, 60, and 70 μg/mL), the fluorescence intensity of the RQDs dramatically decreased (Figure 2D). At high concentrations, the R6 peptide dispersion of RQDs could even be observed directly by the naked eye, since the dispersion showed turbidity. Moreover, the emission wavelength of the RQDs gradually red-shifted from 600 to 605 nm (Figure S4). These results strongly demonstrated that the R6 peptide could neutralize the negative charges of RQDs and trigger quenching through aggregation as a sensitive signal probe.

Double-Enzyme Cascade Amplification. Trypsin, a serine protease possessing endonuclease activity toward peptide chains, hydrolyzed peptide bonds located on the carboxyl side of lysine and arginine residues. Upon incubation of various concentrations of trypsin (0, 5, 10, 15, and 20 μg/mL) with the R6-RQDs (75 μg/mL), significant restoration of fluorescence intensity was observed at 600 nm because trypsin causes the degradation of R6 into smaller fragments or individual arginine molecules (Figure 3A). Compared with those of R6 peptide, the degraded fragments had fewer guanidino groups and positive charges, which did not induce aggregation quenching of the RQDs. Consequently, the

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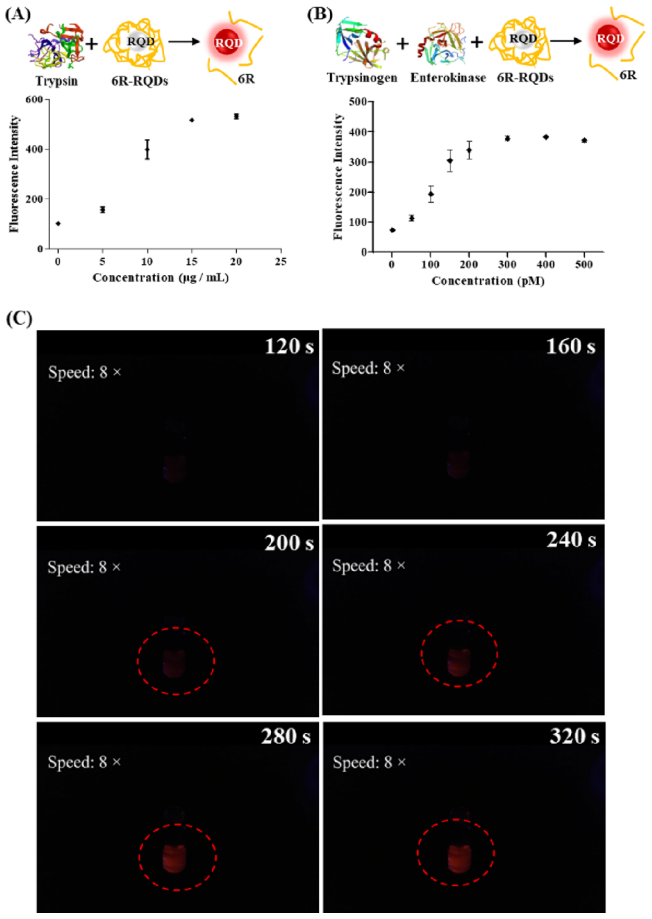


Figure 3. (A) Fluorescence recovery of R6-RQDs (75 μg/mL) by trypsin (0, 5, 10, 15, and 20 μg/mL) and (B) enterokinase-activated trypsinogen. (C) Rapid response time imaging of R6-RQDs (75 μg/mL) to enterokinase (1 nM).

fluorescence of the RQDs gradually recovered with increasing trypsin content. Enterokinase (EK) is a serine protease secreted by mammals and catalyzed the hydrolysis of the carboxy-terminal peptide bond of trypsinogen (TPS) at the N-terminal hexapeptide Val-Asp-Asp-Asp-Asp-Asp-Lys (VDDDDK), resulting in trypsin activation. In general, even a small amount of enterokinase could generate a substantial amount of active trypsin, which restored the fluorescence of the RQDs. As shown in Figure 3B, various concentrations of enterokinase (0, 50, 100, 150, 200, 300, 400, and 500 pM) and trypsinogen were incubated to trigger the fluorescence recovery of the RQDs. Furthermore, when trypsinogen and R6-RQDs were incubated with enterokinase, the red fluorescence intensity of the RQDs rapidly increased after approximately 200 s and eventually reached a plateau at 320 s (Figure 3C). The rapid response was attributed to the efficient signal amplification facilitated by the enterokinase-trypsin dual enzyme system. Moreover, the lowest concentration of R6-RQDs that responded to trypsin was 5 μg/mL, its sensitivity toward enterokinase was significantly enhanced, and the lowest response concentration was low as 50 pM (~1.34 ng/mL). These findings clearly demonstrated the rapid responsiveness of the R6-RQDs to enterokinase within a short time frame (within 5 min), surpassing their sensitivity toward trypsin only. These results highlighted the efficacy of the enterokinase-trypsin double-enzyme signal amplification system.

Method Validation. To confirm the expected process of the enterokinase-trypsin dual enzyme cascade amplification system, SDS-PAGE electrophoresis was conducted to verify the activation of trypsinogen by enterokinase. As shown in Figure 4A, compared with that of trypsinogen (lane 2), the

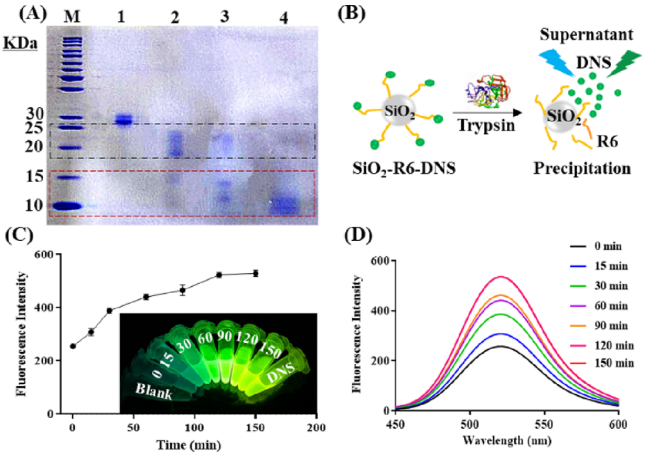


Figure 4. (A)Validation of trypsinogen activation-mediated enterokinase activity by SDS-PAGE (lane 1: enterokinase; lane 2: trypsinogen; lane 3: enterokinase + trypsinogen; lane 4: trypsin). (B) Schematic diagram of trypsin-depredated SiO_2-R6-DNS. (C) Trypsin-catalyzed degradation of SiO_2-R6-DNS (inset photo of the supernatant after incubation of SiO_2-R6-DNS with trypsin for different durations under 365 nm UV light). (D) Changes in the fluorescence intensity of the supernatant at 535 nm after the incubation of SiO_2-R6-DNS NPs with trypsin for different durations (the excitation wavelength was 310 nm).

band near 20−25 kDa of enterokinase-activated trypsinogen (lane 3) exhibited a significant decrease (the black line) in intensity, and a new band appeared near 10−15 kDa (the red line) and corresponded to trypsin (lane 4). This result demonstrated that enterokinase successfully activated trypsinogen and produced trypsin as described previously. To validate the degradation of the R6 peptides by trypsin, the aminated silica nanoparticles (SiO_2) exhibited a spherical-like morphology (Figure S5A). The APTES-modified silica spheres were agglomerated, resulting in an obscured pore structure and confirming successful amino group modification of the silica spheres. The average particle size was approximately 38.99 ± 4.67 nm (Figure S5B). As illustrated in Figure 4B, an N-terminal dansyl chloride-labeled R6 peptide (R6-DNS) was covalently attached to SiO_2 via amide bonding (SiO_2-R6-DNS). The peptide linker connecting the dansyl chloride moiety to the SiO_2 nanoparticles was enzymatically cleaved by trypsin, resulting in detachment of the dansyl chloride from the SiO_2 in the supernatant after centrifugation. As shown in Figure 4C and D, SiO_2-R6-DNS was incubated with trypsin (1 mg/mL) at 37 °C; this resulted in a significant increase in the fluorescence intensity at 535 nm of the supernatant after centrifugation. Conversely, when SiO_2-R6-DNS was incubated with Tris-HCl buffer, minimal enhancement in fluorescence was observed. To further validate the correlation between the enzyme viability of trypsin and its fluorescence recovery effect on R6-RQDs, we employed heat inactivation and phenylmethylsulfonyl fluoride (PMSF), which is a serine protease inhibitor, to inhibit trypsin activity. The fluorescence recovery of R6-RQDs was compared with that of trypsin exhibiting normal viability. As shown in Figure S6, the fluorescence

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intensity was significantly restored upon the addition of active trypsin to the R6-RQDs; this result was consistent with previous experimental findings. Conversely, bath-inactivated trypsin exhibited negligible fluorescence recovery, with an intensity similar to that observed in the negative control group (Tris-HCl). Furthermore, treatment of trypsin with PMSF resulted in a notable reduction in the ability of RQDs to recover their fluorescence (Figure S7), with the reduction becoming more pronounced as the PMSF concentration increased. These results strongly supported that trypsin-mediated hydrolysis of the R6 peptide was responsible for the observed fluorescence recovery phenomenon.

FXIa Inhibitor Detection. To validate the feasibility of FXIa inhibitor detection, we employed FELIAP as a validation model. FELIAP is an aptamer with FXIa inhibitory activity and has been reported to exhibit high affinity for FXIa (KD = 1.8 nM). With this methodology, AptE40 specifically binds to the active site of FXIa during the formation of the MNPs-FXIa-AptE40-QDs-EK complex, acting as a "bridge" that connects the MNPs-FXIa capture probe with the enterokinase-modified AptE40-QDs-EK recognition probe. We assessed both MNPs-FXIa-AptE40-QDs-EK nanocomplexes and enterokinase activity through fluorescent signals. As shown in Figure S8, coincubation of the MNPs-FXIa-QDs-AptE40-EK complex with trypsinogen and R6-RQDs for a specific duration resulted in rapid activation of trypsinogen. The generated trypsin degraded R6 and released the aggregation burst of RQDs, leading to sharp fluorescence recovery of the RQDs. In contrast, the negative control group (Tris-HCl buffer) did not activate trypsinogen and had a low fluorescence intensity at 600 nm. These results confirmed the formation of the MNPs-FXIa-AptE40-QDs-EK complex without affecting enterokinase activity. The FELIAP within the range of 1 to 400 nM caused the release of AptE40-QDs-EK from MNPs-FXIa, leading to the recovery of RQDs fluorescence (Figure 5A) through the enterokinase-trypsin cascade signal amplification system, moreover, a good linear relationship was obtained (fluorescence intensity = 0.3694 [FELIAP] + 437.4, R^2 = 9971) (Figure 5B). Furthermore, precision and accuracy assessments were performed on quality control samples with high (320 nM), medium (180 nM), and low (15 nM) concentrations within the linear range. As shown in Table 1, all concentration levels exhibited accuracies ranging from 91.3% to 101.2%, with relative standard deviations below 15%. The method was highly sensitive with a detection limit of 0.38 nM. In addition, the analysis efficiency was high, and an analysis cycle could be completed within 60 min. These results indicated that the dual enzyme cascade-based signal amplification system had high sensitivity and reliable quantitative performance for detecting the FXIa inhibitors.

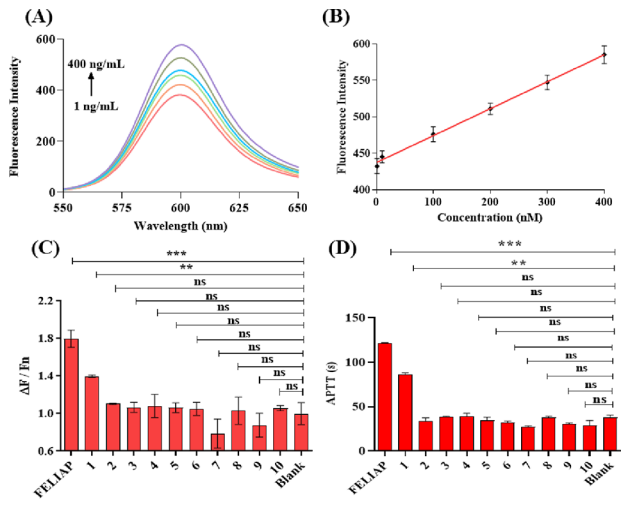


Figure 5. Fluorescence spectra (A) and linearity (B) of RQDs (150 μg/mL) with different concentrations of FELIAP (1, 10, 100, 200, 300, and 400 nM) at 600 nm with an excitation wavelength of 365 nm. (C) FXIa inhibitor screening for crude extracts of *Eupolyphaga sinensis* Walker. (D) Effects of the crude protein extracts from *Eupolyphaga sinensis* Walker on APTT. The data are expressed as the means ± SD, n = 3. Statistical analyses were carried out via one-way ANOVA (ns: $P > 0.05$, **: $P < 0.01$, ***: $P < 0.001$).

Table 1. Accuracy and Precision of FELIAP (n = 5)[a]

spiked conc. (nM)	measured conc. (mean ± SD, nM)	accuracy (%)	RSD (%)
1	1.0 ± 0.1	101.2	9.9
15	13.7 ± 1.7	91.3	12.4
180	177.1 ± 25.6	98.3	14.4
320	318.4 ± 29.8	99.5	9.4

[a]Conc.: concentration. SD: standard deviation. RSD: relative standard deviation. n: number of replicates.

FXIa Inhibitor Screening. The developed method was employed to screen FXIa inhibitors from 10 water-soluble crude protein extracts of *Eupolyphaga sinensis* Walker. As shown in Figure 5C, the detection signal $\Delta F/Fn$ of the positive control compound FELIAP was significantly greater ($p < 0.001$) than that of the negative control (normal saline). Among the 10 extracts of *Eupolyphaga sinensis* Walker, extract no. 1 presented varying degrees of elevated assay signals. Notably, extract no. 1 presented significantly greater assay signals than normal saline ($p < 0.01$); these results indicated the potential presence of higher levels of the FXIa inhibitor. A routine anticoagulant activity investigation also revealed that extract No. 1 significantly prolonged the APTT (Figure 5D); this result significantly differed from that of the blank and other fractions, and no significant prolongation in PT was observed for any fraction (Figure S9). These findings indicated that extract no. 1 effectively extended the APTT by influencing the relevant factors in the endogenous coagulation pathway; these results aligned with the results obtained through our established method. This work provides a crucial technology and analytical approach for screening and detecting FXIa inhibitors in complex extracts, with the potential to be applied in pharmaceutical research.

CONCLUSIONS

In this study, we developed a detection and analysis method based on the fluorescence signal amplification of an enterokinase-trypsin dual-enzyme cascade system. Additionally, the magnetic beads served as solid-phase carriers and provided a larger effective specific surface area to remove other endogenous interfering substances through magnetic separation. Furthermore, this work assessed the feasibility of ultrasensitive analysis in complexes and provided insights into the signal amplification mechanism. The established method provides a key technology for highly sensitive discovery, high-throughput quantitative analysis, and high-content activity screening of the FXIa inhibitors in complexes.

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■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.4c04369.

Materials and reagents; TEM images and particle size distribution of the QDs, RQDs, and SiO_2; UV absorption spectra of the QDs and AptE40-QDs-EK; fluorescence spectra of RQDs with different concentrations of R6 peptide at 600 nm with an excitation wavelength of 365 nm; impact of heat inactivation on the recovery of R6-RQDs fluorescence by trypsin; validation of the enterokinase activity of the MNPs-FXIa-AptE40-QDs-EK complex; effects of the crude protein extracts from *Eupolyphaga sinensis Walker* on PT (PDF)

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Author Contributions

Y.Z. and X.B. performed conceptualization, methodology, software, investigation, formal analysis, and writing−original draft; P.H. performed investigation, software, and supervision; J.N. performed visualization and investigation; Z.Y. conducted investigation; H.C. conducted investigation; Y.D. gathered resources; L.D. gathered resources and performed supervision; C.S. (corresponding author) performed conceptualization, funding acquisition, resource gathering, supervision, and writing−review and editing.

Notes

The authors declare no competing financial interest.

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